Function Notation, Domain, and Range
Answer Key

Algebra 1   Functions

Quick Answers

1. 5	4. 16	7. 20	10. 70
2. 40	5. 7	8. 7.5	11. 72
3. 60	6. 12, 82	9. 264	12. —

Curriculum

Level: Algebra 1   Functions
HSA-CED.A.1–A.4 – *problems 6, 10*
HSF-IF.A–HSF-IF.C – *problems 1, 2, 3, 4, 5, 7, 8, 9, 11, 12*
Difficulty 1–5 of 5 across 14 tagged checks.

Common wrong answers

3. *If they got 4: subtracted the two inputs instead of the two outputs*
4. *If they got 8: squared the negative input as a negative number*
10. *If they got 110: used the first cost in the table as the hourly rate, dropping the fixed launch fee*

1. Evaluate $f(x) = 4x - 7$ at $x = 3$.

Step 1. The notation $f(3)$ names an input, so the only move is to replace every x in the rule with 3.
Step 2. $f(3) = 4(3) - 7 = 12 - 7$.
Step 3. So the output is $\boxed{5}$.
Listen for: a student who answers "f times 3". $f(3)$ is not a product — it is the output of the machine named f when 3 goes in.

2. Read $H(4)$ from Graph B.

Step 1. $H(4)$ asks for the output at the input $t = 4$, so travel along the horizontal axis to 4 and then straight up to the marked point.
Step 2. That point sits at a height of 40 on the vertical axis, and the vertical axis is measured in feet.
Step 3. $H(4) = \boxed{40 \text{ ft}}$.
Teaching note: the drone is at the same height at $t = 2$ and $t = 4$. Two different inputs may share one output — that is allowed for a function. Two different outputs for one input would not be.

3. Compare two readings on Graph B.

Step 1. The question asks how much *higher*, so it is a difference of two *outputs* — read both heights before subtracting anything.
Step 2. From the graph, $H(6) = 100$ and $H(2) = 40$.
Step 3. $H(6) - H(2) = 100 - 40 = \boxed{60 \text{ ft}}$.
Common wrong answer: 4 — the two *inputs* 6 and 2 were subtracted instead of the two heights.

4. Evaluate $g(x) = x^2 - 5x + 2$ at $x = -2$.

Step 1. A negative input is the place substitution goes wrong, so write the parentheses in before simplifying anything: $g(-2) = (-2)^2 - 5(-2) + 2$.
Step 2. $(-2)^2 = 4$ because a negative times a negative is positive, and $-5(-2) = +10$.
Step 3. $g(-2) = 4 + 10 + 2 = \boxed{16}$.
Common wrong answer: 8 — the square was taken as -4, which is what -2^2 means, not what $(-2)^2$ means.

5. Run $f(x) = 4x - 7$ backwards from the output 21.

Step 1. Here the output is known and the input is not, so this is an equation to solve rather than a substitution: $4x - 7 = 21$.
Step 2. Add 7 to both sides: $4x = 28$.
Step 3. Divide by 4: $\boxed{x = 7}$.
Check: $f(7) = 4(7) - 7 = 21$ ✓ — evaluating is how you check a solve, and solving is how you undo an evaluate.

6. Build a rule from Table A, then use it.

(a) Step 1. The problem says the cost rises by the same amount each hour, so the rate is a difference between two neighbouring outputs.
Step 2. $C(2) - C(1) = 34 - 22 = 12$, and the same gap appears again: $46 - 34 = 12$ and $58 - 46 = 12$.
Step 3. The cost rises by $\boxed{12}$ per extra hour.
(b) Step 4. A constant rise of 12 per hour makes the rule linear, $C(x) = 12x + b$, and one table entry pins down b: $22 = 12(1) + b$, so $b = 10$ — a fixed launch fee charged once.
Step 5. The rule is $C(x) = 12x + 10$, so $C(6) = 12(6) + 10 = 72 + 10$.
Step 6. A six-hour rental costs $\boxed{82}$ dollars.
Teaching note: the table never lists 6 hours. Turning the table into a rule is exactly what makes an unlisted input answerable.

7. Where the tank's domain ends.

Step 1. The rule keeps subtracting 24 forever, but the tank cannot hold less than nothing, so the story — not the algebra — decides where the domain stops: at the moment the tank is empty.
Step 2. Empty means $V(t) = 0$, so solve $480 - 24t = 0$, giving $24t = 480$.
Step 3. $t = 480/24 = \boxed{20 \text{ min}}$.
Grading note: full credit needs the reason as well as the number — after 20 minutes the rule would return negative gallons, which the situation has no meaning for. A sensible domain is $0 \leq t \leq 20$.

8. Find the input that gives 300 gallons.

Step 1. The amount of water is the output, so 300 goes on the output side and the unknown minute count is what is being solved for: $480 - 24t = 300$.
Step 2. Subtract 480 from both sides: $-24t = -180$.
Step 3. Divide by -24: $t = 180/24 = \boxed{7.5 \text{ min}}$.
Check: $V(7.5) = 480 - 24(7.5) = 480 - 180 = 300$ ✓. Note that 7.5 minutes is inside the domain found in Problem 7, so it is a legal answer for this story.

9. Evaluate the tank rule at $t = 9$.

Step 1. This time the input is given, so substitute rather than solve: $V(9) = 480 - 24(9)$.
Step 2. $24(9) = 216$, so $V(9) = 480 - 216$.
Step 3. $V(9) = \boxed{264 \text{ gallons}}$, meaning 264 gallons are still in the tank nine minutes after the drain opened.

10. Find and fix the mistake.

Step 1. Test the student's rule against the table before repairing it: $C(x) = 22x$ gives $C(2) = 44$, but Table A says 34. The rule is wrong, so the reasoning behind it is wrong too.
Step 2. What went wrong: 22 is the cost of the *first* hour, not the cost of *each* hour. It already contains the one-time launch fee of 10 dollars, so charging 22 again for every later hour charges that fee over and over.
Step 3. The correct rule separates them: $C(x) = 12x + 10$, so $C(5) = 12(5) + 10 = 60 + 10 = \boxed{70}$ dollars.
Common wrong answer: 110 — the student's rule 22×5.

11. Range and domain for the ticket function.

(a) Step 1. The range is the set of outputs, and $T(n) = 9n$ grows as n grows, so the largest output comes from the largest legal input.
Step 2. The order limit makes 8 the largest legal input, so the top of the range is $T(8) = 9(8)$.
Step 3. The largest value in the range is $\boxed{72}$ dollars.
(b) Step 4. *(Open response — flagged for manual review; read the student's sentence.)* A full-credit answer says both things: tickets come only in whole numbers, so inputs such as 2.5 have no meaning here, and the order limit caps the input at 8. The domain is therefore $\{0, 1, 2, 3, 4, 5, 6, 7, 8\}$ and the range is $\{0, 9, 18, 27, 36, 45, 54, 63, 72\}$ — nine values, not an interval.
Listen for: "the domain is all real numbers, because you can multiply anything by 9." True of the bare rule, false of this situation. The rule and the story together decide the domain.

12. Synthesis — open response, flagged for manual review.

There is no single correct paragraph here. A strong answer contains all of the following.

Drone (Graph B). A realistic domain is $0 \le t \le 8$ seconds — the recorder only stored readings over that window. A realistic range runs from 0 up to 100 feet, the lowest and highest heights the graph reaches. The *graph* makes both easy to see: the domain is how far the picture runs left to right, and the range is how far it runs bottom to top.

Tank ($V(t) = 480 - 24t$). A realistic domain is $0 \le t \le 20$ minutes and the matching range is $0 \le V \le 480$ gallons. The *rule* makes this easier: nothing in a table or a picture announces where the water runs out, but setting the rule equal to 0 finds that instant exactly, and reading the rule at $t = 0$ gives the full 480 gallons.

Why. A graph shows the whole story at once but only to the accuracy the eye can read; a rule is exact and extends past the data, but you must ask it a question to get an answer; a table like Table A is exact at the inputs it lists and silent everywhere else — which is why Problem 6 needed the rule to reach 6 hours.

Grading note: accept any well-argued domain and range, but require the student to name a representation and give a reason. "The graph is easier" alone is not a justification.